

Wayagram – High-Level Product & Technical Architecture Paper

Product: Wayagram – Connect, Share, Shop

Category: Social Media + E-commerce + Fintech Platform

Role Perspective: Technical Product Manager

1. Business Overview (What Wayagram Is)

Wayagram is designed as a **digital super-platform** that unifies social interaction, commerce, and financial services into a single, seamless experience. Unlike traditional platforms that solve only one problem (social media, payments, or e-commerce), Wayagram's value lies in **convergence**.

From a **customer perspective**, Wayagram is a place where users can: - Create an identity and community - Discover content and people - Buy and sell goods and services - Send, receive, and manage money - Promote causes, talents, and events

From a **business perspective**, Wayagram operates as a **multi-sided marketplace** that captures value at multiple interaction points: - Transactions between buyers and sellers - Payments and wallet usage - Advertising and promotions - Platform-hosted events and contests

The platform's strategic advantage is **network effects**: as more users join to socialize, commerce becomes more attractive; as commerce grows, financial services deepen; as money flows, trust and retention increase.

2. Customer Journey (End-to-End)

The Wayagram customer journey is intentionally designed to move users from **discovery** → **engagement** → **transaction** → **loyalty**.

2.1 Discovery & Acquisition

Users discover Wayagram through: - Referral codes (peer-driven growth) - Social sharing of content, events, or marketplace listings - Paid promotions and ads

Technical Enablement: - Referral service generates unique referral codes - Attribution service tracks installs and signups - Reward logic triggers incentives for referrers

2.2 Onboarding & Identity Creation

Once acquired, users go through **self-onboarding**: 1. Account creation (email/phone) 2. Verification (OTP/email) 3. Profile setup (bio, photo, preferences) 4. Optional role upgrade (seller, organizer, advertiser)

Why this matters: Identity is foundational — it determines access, permissions, financial limits, and trust levels.

Technical Flow: - Authentication service validates credentials - Identity service creates user record - Role-Based Access Control (RBAC) assigns permissions

2.3 Engagement & Social Interaction

Users begin interacting with the platform through: - Feed & discovery - Likes, comments, shares - Pages, groups, and following - Chat, voice, and video

This phase builds **habit and stickiness**.

Technical Considerations: - Feed ranking algorithms - Real-time messaging infrastructure - Media storage and CDN delivery

2.4 Monetization & Transactions

As trust builds, users transact: - Fund wallets via virtual accounts or cards - Buy and sell on the marketplace - Donate to causes - Pay for votes, promotions, and ads

Money movement is the **highest-risk and highest-value** journey.

2.5 Retention, Trust & Support

Long-term retention is driven by: - Notifications - Analytics dashboards - Dispute resolution - Transparent transaction history

Trust is reinforced through visibility, logs, and responsive support.

2.2 High-Level Customer Journey

1. **Discovery & Acquisition**
2. App download / Web access
3. Referral codes
4. Ads & shared content

5. Onboarding

- 6. Account creation (self-onboarding)
- 7. Profile setup
- 8. Role assignment (user, seller, admin)

9. Engagement

- 10. Feed, discovery, moments
- 11. Likes, comments, sharing
- 12. Chat, calls, live streaming

13. Monetization & Transactions

- 14. Wallet funding (virtual accounts, cards)
- 15. E-commerce purchases
- 16. Bidding & quotes
- 17. Donations & voting payments

18. Retention & Growth

- 19. Notifications
- 20. Promotions & ads
- 21. Analytics dashboards

22. Support & Trust

- 23. Dispute resolution
- 24. Audit logs
- 25. Compliance & security

3. Business Journey (Value Creation)

Revenue Streams

- Transaction fees (wallet, payments)
- Marketplace commissions
- Advertising & promotions
- Featured listings
- Event hosting fees
- Enterprise / brand pages

Business KPIs

- Monthly Active Users (MAU)
 - Transaction success rate
 - GMV (Gross Merchandise Value)
 - Wallet balances & velocity
 - Conversion & retention
 - Ad engagement
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4. High-Level System Architecture

4.1 Architectural Principles

- Modular microservices
 - Event-driven architecture
 - Secure financial ledger
 - Horizontal scalability
 - High availability & fault tolerance
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4.2 Core System Layers

1. Client Layer - Mobile Apps (iOS / Android) - Web Application

2. API Gateway - Authentication - Rate limiting - Request routing

3. Core Services Layer - User & Identity Service - Social Graph Service - Content & Feed Service - Messaging & Realtime Service - Wallet & Ledger Service - Payments & Settlement Service - Marketplace Service - Ads & Promotion Service - Analytics & Reporting Service

4. Data Layer - Relational DB (transactions, users) - NoSQL DB (feeds, chats) - Object Storage (media, files) - Cache (Redis)

5. Infrastructure Layer - Cloud (AWS / GCP / Azure) - Kubernetes / Containers - CI/CD pipelines - Monitoring & logging

5. Feature-by-Feature Technical Breakdown

This section explains **how each major feature is implemented technically**, including system responsibilities and data flow.

5.1 Account Creation, Roles & Access

Purpose: Establish identity, permissions, and trust boundaries.

Engineering Approach: - Centralized Identity Service - Role-Based Access Control (RBAC)

Process Flow: 1. User submits signup request 2. Verification service validates identity 3. User profile is created 4. Default role assigned (basic user) 5. Additional roles added as needed (seller, admin)

Why it matters: Prevents unauthorized actions and supports scalable permission management.

5.2 Wallet, Virtual Accounts & Payments

Purpose: Enable secure money storage and movement.

Core Principle: Every financial action must be traceable, reversible, and auditable.

Engineering Approach: - Dedicated Wallet Service - Immutable Ledger Service - Payment Gateway Integrations

High-Level Flow: - Virtual account assigned to user - Incoming transfer triggers webhook - Ledger records credit - Wallet balance updated

Double-Entry Ledger: - Every credit has a corresponding debit - Ensures reconciliation accuracy

5.3 Marketplace & E-Commerce

Purpose: Enable buying and selling within the platform.

Engineering Components: - Product Catalog Service - Order Management Service - Pricing & Commission Engine

Process Flow: 1. Seller lists product 2. Buyer places order 3. Payment processed 4. Funds held in escrow 5. Seller paid after fulfillment

This protects buyers and builds trust.

5.4 E-Bidding System

Purpose: Enable demand-driven commerce.

Flow: - Buyer posts request - Sellers submit quotes - Buyer selects offer - Payment and fulfillment follow

This introduces price discovery and competition.

5.5 Donations, Voting & Talent Programs

Purpose: Support social impact and engagement monetization.

Engineering Considerations: - Real-time counters - Fraud detection - Rate limiting

Votes and donations are treated as **financial transactions**.

5.6 Social Content, Feed & Discovery

Purpose: Drive engagement and discovery.

Architecture: - Event-driven feed generation - Hybrid fan-out model

This balances performance and personalization.

5.7 Chat, Calls & Live Streaming

Purpose: Real-time communication.

Technology: - WebSockets for chat - WebRTC for voice/video - Media servers for live streaming

Scalability and latency are key.

5.8 Ads, Promotions & Analytics

Purpose: Monetization and growth.

Flow: - Campaign setup - Targeting rules - Impression and conversion tracking

5.9 Notifications, Logs & Dispute Resolution

Purpose: Trust, transparency, and support.

Engineering Approach: - Centralized audit logging - Ticketing and escalation workflows

6. Data, Security & Compliance

- Encryption at rest & transit
- Role-based access

- Audit trails
 - Webhook verification
 - PCI-DSS alignment
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7. Scalability & Reliability

- Horizontal scaling
 - Circuit breakers
 - Retry & idempotency
 - Graceful degradation
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8. Metrics & Monitoring

- Transaction success rate
 - Feed latency
 - Wallet reconciliation gaps
 - Fraud alerts
 - System uptime
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9. Roadmap Strategy (High Level)

Phase 1: Core social + wallet + marketplace

Phase 2: Ads, bidding, donations, analytics

Phase 3: Live streaming, talent programs, enterprise tools

10. Final Product Vision

Wayagram is positioned to become a **digital super-platform** where social interaction, commerce, and finance coexist seamlessly — optimized for scale, trust, and monetization.

As a Technical Product Manager, success lies in **system reliability, transaction integrity, and user trust** while enabling rapid feature expansion.

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